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Prediction of traffic congestion through Decision Tree, Queuing Theory, Deep Belief Network and Random Forest methods

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Abstract
The increase in vehicle ownership and urban population puts a strain on roads and increases traffic congestion within developed and developing nations. Congestion affects an individual's time and increases an individual's fuel consumption and pollution output, undercutting efforts towards smart/ sustainable cities. Outdated statistical congestion predicting methods do not account for the non-linear, dynamic nature of congestion with regards to external factors such as, weather, road incidents, and infrastructure. The adoption of machine and deep learning develops the framework towards adaptable and data-driven congestion predicting methods. This study develops a framework that combines the fundamentals of machine learning algorithms, Queuing Theory, Decision Trees, Random Forests, and Deep Belief Networks (DBN) for robust and interprefig traffic congestion forecasting. The traffic flow micro-dynamics integration with Queuing Theory and the use of historical/ real-time multimodal data such as sensor feeds, GPS trajectories, and incident reports as a basis for forecasting helps in robust congestion prediction. Polynomials of moving and rest queues with defined traffic characteristics fuel flow mechanisms. The designed framework uses the DBN and Queuing Theory for robust traffic forecasting.

Keywords Traffic congestion, Machine learning, Decision Tree, Random Forest, Deep Belief Network, Queuing Theory, Intelligent transportation system, Traffic prediction, Data analytics, Hybrid model, Urban mobility

1 Introduction

1.1 Background and motivation

Rapid urbanization and increasing vehicle ownership have made traffic congestion a persistent challenge in modern cities, impacting travel reliability, economic productivity, environmental sustainability, and public health [1, 2]. Effective congestion prediction is essential for transportation planning, resource allocation, and real-time traffic management [3], particularly in large metropolitan areas [4].

Traffic congestion arises from complex interactions among temporal demand variations, spatial network structure, weather conditions, incidents, construction activities,

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and driver behaviour [5–7]. Traditional statistical models, such as ARIMA, struggle to capture these nonlinear and dynamic relationships [8, 9], despite the growing availability of real-time data from GPS, sensors, and cameras [10, 11].

While recent deep learning and graph-based approaches have shown promising predictive performance [7, 12], they often rely on dense sensing infrastructure and black-box architectures, limiting interpretability and deployment flexibility in heterogeneous urban environments [5, 13]. These challenges motivate the development of hybrid frameworks that combine interpretable traffic modelling with data-driven learning to enable accurate and deployable congestion prediction.

1.2 Research gap and problem formulation

Despite significant advances in traffic congestion prediction, several limitations remain. Many existing models rely heavily on historical data and lack adaptability to sudden environmental changes such as weather disruptions, road conditions, or incidents [6, 14]. In addition, incomplete integration of heterogeneous data sources—such as GPS, traffic sensors, weather, and contextual information—restricts comprehensive spatio-temporal modelling [6, 7, 15].

Recent graph-based deep learning models often require explicit graph construction and dense sensor deployments, which can limit their applicability in real-world urban settings [5, 13]. Conversely, classical traffic flow models offer interpretability but insufficient predictive power under highly dynamic conditions. These gaps motivate a hybrid framework that balances interpretability, predictive performance, and practical deployment feasibility.

1.3 Research problem

Despite significant progress in traffic prediction research, several gaps still remain in current approaches:

1. Limited Handling of Dynamic Environmental Changes: Many existing models depend strongly on past traffic patterns and do not adapt well to sudden external events such as weather fluctuations, road damage, or unexpected incidents [6, 14]. This limits their reliability in rapidly changing real-world conditions.
2. Incomplete Multimodal Data Integration: Only a limited number of methods successfully combine diverse data sources such as traffic sensors, GPS signals, weather information, and social media inputs. Without effective integration of these heterogeneous datasets, models cannot fully represent the spatial and contextual complexity of urban mobility systems [6, 7, 15].
3. Scalability and Interpretability: Deep learning approaches often achieve high prediction accuracy but tend to be computationally demanding and difficult to interpret. On the other hand, ensemble techniques and queuing-based models offer clearer explanations but may struggle with highly nonlinear traffic behaviour [6, 8, 16].
4. Inadequate Attention to Real-Time and Short-Term Prediction: Some existing frameworks focus mainly on limited forecasting horizons or rely on centralized architectures that restrict real-time deployment. These limitations reduce their effectiveness in large-scale, time-sensitive traffic applications [10, 17, 18].

1.4 Research contribution

In this paper, we address the limitations of existing traffic congestion prediction approaches by proposing a hierarchical hybrid framework that integrates queuing theory with Decision Tree, Random Forest, and Deep Belief Network (DBN) models. The key contributions of this work are summarized as follows:

1. Hierarchical Integration Strategy: Unlike existing hybrid approaches that combine machine learning and deep learning models in parallel or for benchmarking purposes, the proposed framework introduces a structured decision–ensemble–deep learning pipeline, enabling controlled information flow from interpretable models to deep representation learning.
2. Queuing-Theoretic Deviation Modelling: Queuing theory is not used merely as a standalone analytical tool, but is explicitly employed to model real-time traffic state deviations, which are injected into the learning pipeline to enhance responsiveness to non-recurrent congestion events.
3. Interpretability-Guided Learning: Decision Trees provide rule-based congestion characterization and dominance selection, while Random Forests stabilize predictions and refine feature representations before deep learning is applied.
4. Deep Spatiotemporal Pattern Learning: The DBN operates on refined ensemble outputs rather than raw traffic data, enabling effective learning of nonlinear temporal and spatial congestion patterns while maintaining system-level interpretability.
5. Comprehensive Empirical Validation**:** The proposed framework is validated on multiple real-world urban traffic datasets, demonstrating consistent improvements over classical statistical models and conventional machine learning approaches.

1.5 Related work and positioning

Recent graph neural network (GNN)–based approaches model spatial dependencies in road networks and have shown strong predictive performance under dense sensor deployments. However, such models typically rely on end-to-end deep learning, explicit graph construction, dense sensing infrastructure, and high computational cost, which limits their applicability in heterogeneous or data-sparse urban settings.

In contrast, the proposed framework adopts a modular, graph-aware strategy, where spatial relationships are incorporated during preprocessing, while prediction is performed using interpretable machine learning and hierarchical deep learning components. This design avoids the complexity of full end-to-end GNN training while retaining spatial awareness.

Recent physics-informed and graph-attention-based models, including physics-informed graph attention transformers and Bayesian ensemble graph attention networks, have demonstrated promising results in traffic and air traffic prediction [19, 20]. Data-driven governing equation identification methods have also been explored for modelling traffic flow dynamics [21]. While effective, these approaches emphasize model complexity and end-to-end learning. In comparison, the proposed framework adopts a queue-aware hierarchical design that balances interpretability, scalability, and predictive performance without requiring fully end-to-end deep architectures.

As summarized in Table 1, existing approaches primarily emphasize standalone deep spatio-temporal modeling or ensemble accuracy improvements. In contrast, the

Table 1 Comparison of representative traffic congestion prediction approaches and their methodological characteristics

Category	References	Core methodology	Integration	Interpretability	Architecture
Classical statistical baseline	Yu et al. [22]	High-order markov chain modeling with probabilistic state transitions	Standalone stochastic forecasting	High (explicit probabilistic structure)	Statistical time-series prediction
Deep spatio-temporal neural networks	Song et al. [1]; Yu et al. [3]	CNN-based and attention-based spatio-temporal neural architectures	end-to-end deep learning	Low–moderate (black-box models)	Joint spatial–temporal feature learning
Recurrent neural network models	Ma et al. [23]; Lv et al. [16]	LSTM-based sequence modeling (including hybrid denoising variants)	Deep learning with temporal memory	Low–moderate	Temporal dynamics modeling
Ensemble learning approaches	Wang et al. [17]; Zhang et al. [14]	Consensus and hybrid ensemble prediction frameworks	multi-model integration	Moderate (component models interpretable)	Ensemble-based traffic prediction
Proposed hybrid framework (this work)	NA	Queuing-theoretic features + decision tree + random forest + deep belief network	Structured hybrid architecture	Moderate–high (interpretable components + deep learning)	Theory-guided hierarchical learning

proposed framework integrates queueing-theoretic features with interpretable ensemble learning and hierarchical deep representation within a unified pipeline, addressing interpretability and robustness alongside predictive performance.

While several hybrid frameworks integrate theoretical traffic models with deep learning by embedding physical constraints directly within end-to-end neural networks (e.g., queueing-theory-guided LSTM models and physics-informed neural architectures), these approaches typically couple theory and learning within a single monolithic model. In contrast, the proposed framework adopts a staged hierarchical integration strategy in which queueing-theoretic deviation modeling is performed as an external analytical layer and subsequently propagated through an interpretable ensemble module before deep representation learning. This architectural separation enables modular interpretability, flexible deployment, and targeted modeling of congestion dominance patterns. Therefore, the contribution of this work lies not in embedding theoretical constraints inside a neural network, but in formalizing a multi-layer integration pipeline that explicitly organizes theoretical reasoning, ensemble decision learning, and deep hierarchical modeling into a unified predictive architecture.

2 Literature review

2.1 Classical and statistical models

Early studies on traffic prediction mainly relied on classical statistical techniques such as ARIMA (Autoregressive Integrated Moving Average). These models worked reasonably well for simple, low-dimensional time-series data [1, 24, 25]. However, real urban traffic is highly dynamic and nonlinear. Sudden disruptions and irregular patterns expose the limits of purely statistical methods [8, 9, 26, 27]. Traditional regression approaches also struggle to include external factors such as weather or holidays. In addition, they are not well suited for capturing complex spatial and temporal relationships in large traffic networks [6, 18].

2.2 Machine learning approaches

To overcome these limitations, researchers began exploring non-parametric machine learning methods. Common examples include k-Nearest Neighbours (KNN), Support Vector Machines (SVM), Bayesian Networks, and Artificial Neural Networks (ANN) [3, 10, 28, 29]. Kernel-based SVM models can describe nonlinear traffic patterns more effectively than classical methods [6, 9, 10]. Even so, many of these techniques have difficulty modelling long-term spatial and temporal interactions. They may also face challenges when selecting useful features from high-dimensional traffic data.

Random Forests represent another important development in this area. As an ensemble of Decision Trees, they are robust to noise and missing values and help reduce overfitting [6, 7]. They can estimate feature importance and handle both categorical and numerical variables. These properties make them practical for traffic congestion prediction tasks [6, 7].

2.3 Deep learning and hybrid models

Deep learning has significantly changed modern traffic prediction. These models can learn complex patterns from large and diverse datasets. Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) networks, are widely used for modelling sequential traffic behaviour [7, 23, 30, 31, 36]. Convolutional Neural Networks (CNNs) help capture spatial patterns and local congestion structures in road networks [12, 13, 22, 32]. However, training large deep architectures, such as Factorization Machines or Deep Belief Networks, can require substantial computational resources, particularly when high-resolution inputs are used [8, 15, 33].

To balance performance and interpretability, many recent studies combine learning methods with domain knowledge. Hybrid models integrate machine or deep learning with theoretical frameworks such as queuing theory or graph theory [6, 7, 16, 34]. For instance, combining graph neural networks with traditional ML classifiers helps represent complex spatial relationships in large traffic systems [5, 14, 35].

2.4 Data-driven versus model-driven approaches

Traffic prediction research often distinguishes between model-driven and data-driven strategies [6]. Model-driven approaches are based on theoretical principles, such as queuing theory. They provide clear and interpretable insights but may lack flexibility in changing environments. In contrast, data-driven methods focus on learning patterns directly from data. These methods adapt well to noisy and complex datasets, yet they can be less transparent and usually require large labelled datasets for training [7, 15].

2.5 Multimodal fusion and real-world application

Recent studies show that combining multiple data sources can significantly improve traffic prediction. Researchers increasingly use multimodal fusion, where traffic data is integrated with weather information, incident reports, and even social media signals [6, 15, 36, 37]. This combination helps models become more accurate and stable under changing conditions. Several deep learning frameworks have been developed to learn shared representations from different data types, including multimodal CNNs and Deep Boltzmann Machines [15, 16].

Real-world implementations further highlight the need for practical and scalable systems. Public datasets such as the IBB Traffic Graph dataset [5] and the PeMS system in California [6, 7] demonstrate that modern traffic prediction requires solutions that operate in real time and can scale across large networks.

3 Methodology

3.1 Overview of the proposed framework

The proposed framework combines theoretical modelling with modern learning techniques. It bridges model-driven and data-driven approaches within a single architecture. At its foundation is a queuing-theoretic model that represents vehicle flow dynamics. On top of this layer, a hierarchical learning pipeline is built using Decision Tree, Random Forest, and Deep Belief Network (DBN) components.

The framework processes both historical and real-time multimodal inputs. These include GPS trajectories, traffic sensor counts, weather variables, and incident information (see Table 1). By integrating these diverse data sources, the system aims to produce accurate and interpretable congestion predictions attribute along with description as given in Table 2.

The framework's operation is divided into two primary phases: Realtime Learning (for immediate adaptation and prediction) and Model Learning (for continuous improvement and retraining).

It is important to emphasize that the proposed framework does not merely benchmark multiple prediction models independently. Instead, it defines a hierarchical information flow, where queuing-theoretic traffic state deviations guide rule-based and ensemble learning stages, whose outputs are subsequently used as structured inputs to the Deep Belief Network. This integration strategy enables the framework to jointly achieve interpretability, robustness, and predictive accuracy.

3.2 Queuing theory-based traffic modelling

3.2.1 Theoretical foundation

Queuing theory provides a formal mathematical framework for modelling the flow and accumulation of vehicles through a network of nodes (road segments, intersections) [11, 17]. Each service node represents a spatial region (e.g., a road or intersection), where vehicles arrive at a certain rate (λ), are serviced (i.e., pass through) at another rate (μ), and may queue if demand exceeds capacity.

The queuing models considered in this work, including M/M/1 and M/D/1 formulations, are employed as analytical approximations to characterize the imbalance between traffic arrival and service processes under congestion. While real-world traffic systems exhibit non-stationary and heterogeneous behaviour, such models provide a tractable and interpretable representation that has been widely adopted in transportation and traffic

Table 2 External contextual factors considered for traffic congestion modelling

Attribute	Description
Weather	Rain, temperature, humidity, etc
Time	Hour, day of week, season
Holiday	Boolean event indicator
Road quality	Pavement, deformation, construction
VIP movement	Incidents, processions
Construction work	Ongoing roadwork status

flow analysis. Importantly, the proposed framework does not rely on strict assumptions of a specific queuing formulation; instead, queuing-theoretic state deviations are used as relative indicators of congestion severity. This design allows the framework to be readily extended to more general or data-driven queuing models when required.

3.2.2 Queuing metrics

1. Arrival Rate (λ): Frequency of vehicles entering a node, estimated from GPS or sensor data.
2. Service Rate (μ): Frequency of vehicles exiting or being processed by a node.
3. Queue Length (Lq): Number of vehicles queued at a node.
4. Waiting Time (Wq): Average delay experienced by vehicles before service.

Congestion is formally identified when the traffic arrival rate exceeds the service rate:

$$\lambda > \mu \tag{1}$$

Resulting in queue buildup and increased delays. Real-time estimation of these metrics is achieved by mapping GPS trajectories to spatial nodes and tracking time-stamped vehicle entries and exits [11, 17].

3.2.3 Spatial and temporal data mapping

1. Let $A = \{a_1, a_2, \ldots, a_n\}$ denote the set of spatial nodes (road segments or intersections).
2. Let $G = \{g_i(t)\}$ denote GPS observations, where $g_i(t)$ represents the location of vehicle i at time t.

Each GPS data point is mapped to its nearest spatial node, and temporal interpolation is used to estimate mid-segment positions and movement. This enables the construction of virtual arrival/departure events and supports real-time queue estimation.

3.2.4 System estimation and deviation

The system deviation is defined as the difference between the observed and expected traffic states, denoted by.

$$HP(t) = h_p(t) - \widehat{h}_p(t) \tag{2}$$

where, $h_p(t)$ denotes the observed traffic state at time t, and

$\widehat{h}_p(t)$ represents the expected traffic state estimated using queuing theory.

It captures real-time anomalies in service performance and serves as an input to the learning pipeline.

For notational clarity, traffic flow observations over a finite temporal window of length m are represented as a time-indexed sequence.

$$X(t) = \{f(t), f(t-1), \ldots, f(t-m+1)\} \tag{3}$$

where, $f(t)$ denotes the traffic flow at time t.

3.2.5 Batch processing and temporal segmentation

To facilitate efficient computation, spatiotemporal data is partitioned into observation intervals (chunks). Queuing metrics (e.g., using Little's Law $L = \lambda W$, where L is the

average queue length, λ is the arrival rate, and W is the average waiting time.) are computed over each interval, supporting dynamic arrival/service rate estimation and temporal bottleneck identification.

3.2.6 Notation summary

For clarity, the main symbols used in the queuing-theoretic modelling and learning framework are summarized in Table 3.

3.2.7 Sensitivity of queuing parameters

In the proposed framework, queuing-theoretic parameters—arrival rate (λ) and service rate (μ)—are not used for exact stochastic traffic simulation but as relative indicators of congestion deviation. The condition $\lambda > \mu$ reflects congestion buildup, while the magnitude of deviation guides downstream learning components.

To evaluate robustness, λ and μ were varied within realistic bounds derived from empirical traffic observations (± 10–20%). The experimental results indicate that moderate variations in these parameters do not significantly affect prediction accuracy, as the hierarchical learning layers adaptively absorb such fluctuations. This demonstrates that the framework is robust to parameter uncertainty and suitable for real-world deployments where precise queuing parameters may be noisy or approximate.

The queuing-theoretic indicators derived in this section are subsequently used as structured inputs to the learning components of the proposed model, thereby providing a theoretical foundation for guiding hierarchical traffic congestion prediction.

3.3 Machine learning and deep learning algorithms

3.3.1 Decision Tree

In our proposed framework, Decision Tree has been used to perform initial rule-based classification and to determine the dominance of real-time versus historical traffic data under different operating conditions [6, 7].

3.3.2 Random Forest

Random Forest is employed to enhance robustness, handle missing values, and perform feature importance estimation within the hierarchical learning pipeline [6, 7].

Table 3 Summary of symbols used in the queuing-theoretic and learning framework

Symbol	Description
λ	Traffic arrival rate at a node
μ	Traffic service rate
L	Average queue length
W	Average waiting time of vehicles
$f(t)$	Traffic flow at time t
$X(t)$	Temporal traffic feature sequence over a window of length m
m	Temporal observation window length
t	Time index
$h_p(t)$	Observed traffic state at time t
$\widehat{h}_p(t)$	Expected traffic state estimated using queuing theory
$HP(t)$	Traffic state deviation between observed and expected states
h	Hidden unit state in an RBM
$H^{(l)}$	Hidden-layer representation produced by the -th RBM

3.3.3 Deep Belief Network (DBN)

The Deep Belief Network (DBN) is used to capture nonlinear spatiotemporal patterns from the fused traffic features, supporting short-term and multi-step congestion prediction [15, 16, 38].

3.3.4 DBN Architecture and training protocol

The Deep Belief Network (DBN) in the proposed framework is designed to learn hierarchical representations from aggregated traffic features produced by the Decision Tree and Random Forest layers. The DBN consists of multiple stacked Restricted Boltzmann Machines (RBMs), where each RBM learns progressively higher-level abstractions of traffic patterns.

1. Architecture:The DBN architecture comprises an input (visible) layer followed by multiple hidden layers, with each hidden layer corresponding to one RBM. The number of hidden layers and units per layer is selected empirically based on validation performance to balance representational capacity and computational efficiency. The output of each RBM serves as the input to the subsequent layer, enabling hierarchical feature learning.
2. Training Protocol:Training is performed using a two-stage procedure. First, unsupervised layer-wise pre-training is conducted for each RBM using Contrastive Divergence to initialize network parameters and stabilize learning. This is followed by supervised fine-tuning of the entire network using labelled congestion states to optimize prediction performance. A fixed learning rate is employed during training, and early stopping based on validation loss is used to prevent overfitting.
3. Convergence Behaviour:The unsupervised pre-training stage ensures stable convergence by learning meaningful feature representations before supervised optimization. Empirically, the training loss stabilizes within a limited number of epochs, indicating convergence of the DBN parameters. This hierarchical learning strategy enables the model to capture both short-term fluctuations and longer-term temporal dependencies in traffic dynamics without relying on explicit recurrent connections.

Modelling Long-Range Temporal Dependencies:

The proposed framework captures long-range temporal dependencies through a combination of temporal segmentation and hierarchical representation learning rather than explicit recurrent connections. Traffic observations are first aggregated into temporal windows, allowing historical context beyond instantaneous measurements to be incorporated into the feature space. These temporally structured inputs are then processed by the Deep Belief Network, where stacked Restricted Boltzmann Machines learn hierarchical representations over successive layers.

This layer-wise abstraction enables the DBN to encode both short-term fluctuations and longer-term temporal patterns in traffic dynamics. In addition, queuing-theoretic deviation features aggregated over temporal windows provide complementary contextual signals that further enhance the model's ability to respond to evolving congestion patterns over extended time horizons.

3.3.5 Multimodal data fusion

A key aspect of the proposed framework is the integration of multiple data sources to improve prediction quality [6, 7, 15]:

1. Sensor and GPS Data: Real-time information on vehicle counts, speed, and movement patterns.
2. Weather Data: Environmental factors such as temperature, rainfall, humidity, and wind speed.
3. Incident Reports: Records of accidents, road work, and temporary closures.
4. Social media and Points of Interest: Signals related to events that may cause sudden traffic demand.

Feature engineering techniques convert raw inputs into structured representations suitable for model training. Examples include traffic density measures, time-based lag features, and spatial encodings. For graph-structured traffic networks, node embedding methods such as GLEE and Node2Vec are applied to represent relationships between connected road segments [5].

3.3.6 Model training and evaluation

The model is trained using both historical records and live traffic data under standard supervised learning procedures. Its performance is assessed through commonly used evaluation metrics, including precision, recall, F1-score, Area Under the ROC Curve (AUROC), and RMSE. These measures allow the framework to be evaluated across both classification and regression tasks [5–7].

4 Algorithms for proposed model

The proposed framework integrates queuing theory with Decision Tree, Random Forest, and Deep Belief Network (DBN) models for traffic congestion prediction. Queuing theory provides a structured mathematical basis for modelling traffic dynamics such as arrival rates, service times, queue lengths, and delays, enabling characterization of current traffic conditions. To support forecasting and proactive decision-making, this analytical foundation is combined with machine learning models that learn from historical traffic patterns [5, 14].

Decision Trees generate interpretable, rule-based congestion classifications using queue-derived and contextual features. Random Forests enhance robustness by aggregating multiple trees trained on feature subsets, capturing complex relationships between traffic indicators and congestion states while reducing overfitting.

To model longer-term and nonlinear temporal dependencies, a Deep Belief Network is employed. Features derived from the queuing model and ensemble layers are processed through multiple hidden layers, enabling the extraction of hierarchical temporal patterns and improving sensitivity to abrupt traffic changes caused by external events.

By integrating theory-driven traffic modelling, interpretable ensemble learning, and deep temporal representation learning, the proposed hybrid framework achieves a balanced trade-off between interpretability, robustness, and predictive performance, resulting in a scalable solution for intelligent traffic congestion prediction [5, 14].

4.1 Decision tree for traffic congestion

The Decision Tree module is employed for initial rule-based congestion classification and for determining the dominance between real-time and historical traffic inputs. The model operates on queuing-derived and contextual features and produces an interpretable decision path used to guide downstream learning component.

Figure 1 illustrates the interpretable decision hierarchy used for congestion classification. The model first evaluates average traffic speed as a primary indicator of system state.

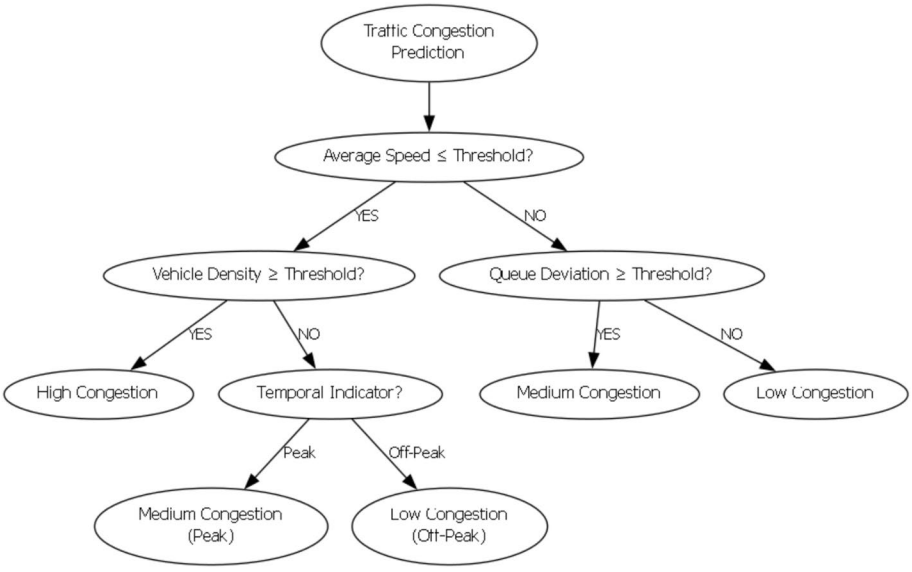


Fig. 1 Illustrates the rule-based decision structure used for the initial assessment of traffic congestion, integrating contextual and aggregated traffic features to classify congestion states

Subsequent decision nodes incorporate vehicle density and queue deviation measures to distinguish between high, medium, and low congestion regimes. Temporal indicators further refine intermediate congestion states by accounting for peak and off-peak traffic patterns. This hierarchical structure enables transparent mapping from aggregated traffic features to discrete congestion classes while preserving interpretability of the decision process.

Algorithm 1 Tree_creation
(Sample:S, Feature:F, DecisionNode:D)

Initialize Decision Tree: DT = BuildTree(S, F, d)
Return DT

Function BuildTree(S, F, d)
If all examples in S have the same congestion level, return a leaf node with that level
If F is empty or d = 0, return a leaf node with the majority congestion level in S
For each feature f in F, do
  Calculate Information Gain (IG) for f based on congestion levels
End For
Select the feature f* with the highest IG
Create a decision node with feature f*
For each possible value v of f*, do
  S_v = subset of S where f* = v
  If S_v is empty, then
    Add a leaf node with the majority congestion level in S
  Else
    F_v = F - {f*}
    child = BuildTree(S_v, F_v, d-1)
    Add child as a branch of the decision node
  End If
End For
Return the decision node
End Function

4.2 Creation of Random Forest for traffic congestion

The Random Forest module enhances robustness by aggregating multiple decision trees trained on random feature subsets. In the proposed framework, it is primarily used for handling missing data, reducing variance, and estimating feature importance prior to deep learning.

Figure 2 presents the ensemble learning structure of the Random Forest model. Training traffic data are processed by multiple decision trees operating in parallel, each learning complementary decision patterns from the feature space. The outputs of these trees are combined using a majority-voting strategy, which reduces prediction variance and improves robustness. The aggregated decision yields the final congestion classification into low, medium, or high traffic states. This ensemble mechanism enhances predictive stability compared to a single decision tree while preserving interpretability.

Algorithm 2 Creation_Random Forest (trees T)

```
Initialize the Random Forest with an empty list of trees: RF = []
For t = 1 to T do
   Draw a bootstrap sample StS_t with replacement from dataset S
   Build a decision tree Tree_t = BuildTree(S_t, F, d, m)
   Add Tree_t to RF
End For
Return RF

Function BuildTree(S, F, d, m)
   If all examples in S have the same class, return a leaf node with that class
   If F is empty or d = 0, return a leaf node with the majority class in S
   Randomly select m features from F to form a feature subset
   For each feature f in the selected subset do
      Calculate Information Gain (IG) for feature f
   End For
   Select the feature f* with the highest Information Gain
   Create a decision node with feature f*
   For each possible value v of feature f* do
      Let S_v be the subset of S where f* = v
      If S_v is empty then
         Add a leaf node with the majority class in S
      Else
         Let F_v = F - {f*}
         child = BuildTree(S_v, F_v, d - 1, m)
         Add child as a branch of the decision node
      End If
   End For
   Return the decision node
End Function
```

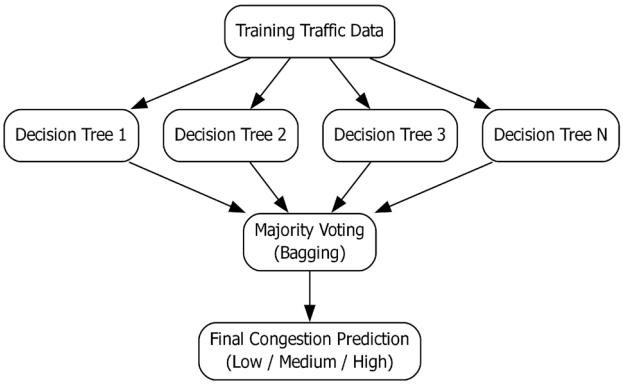


Fig. 2 Illustrates the Random Forest ensemble framework used for traffic congestion prediction. Multiple decision trees are trained on the same traffic dataset, and their individual predictions are aggregated through a majority-voting (bagging) mechanism to produce the final congestion classification

4.3 Deep Belief Network for traffic congestion

The DBN module captures nonlinear spatiotemporal dependencies in the fused traffic features. It is responsible for learning latent representations that support short-term and multi-step congestion prediction under dynamic traffic conditions.

Algorithm 3 Deep belief network training

Input: X (input feature vector)

Output: $H^{(l)}$ (final hidden-layer representation)

Parameters: Weights W, biases a (visible), b (hidden), learning rate η

Here, $H^{(l)}$denotes the hidden-layer representation produced by the l-th RBM, while $h$represents individual hidden units within an RBM.

Initialize all weight matrices W^l and bias vectors a^l, b^l (for layers l = 1 to L) randomly
Set input for first layer: $H^0 = S$

For l = 1 to L (each RBM layer in the DBN), do
Use H^{l-1} as input to RBM layer l
For each training sample $x \in H^{l-1}$, do
• Initialize visible vector: $v^0 \leftarrow x$
• For t = 0 to k–1, do
– Sample h^t from $P(h \mid v^t)$
– Sample v^{t+1} from $P(v \mid h^t)$

For each pair (i, j) of visible and hidden units
– $\Delta W^{(l)}_{ij} = \eta \times [P(h_j = 1 \mid v^0) \times v^0_i - P(h_j = 1 \mid v^k) \times v^k_i]$
– $W^{(l)}_{ij} \leftarrow W^{(l)}_{ij} + \Delta W^{(l)}_{ij}$

– $\Delta a^{(l)}_j = \eta \times [P(h_j = 1 \mid v^0) - P(h_j = 1 \mid v^k)]$
– $a^{(l)}_j \leftarrow a^{(l)}_j + \Delta a^{(l)}_j$

– $\Delta b^{(l)}_i = \eta \times (v^0_i - v^k_i)$
– $b^{(l)}_i \leftarrow b^{(l)}_i + \Delta b^{(l)}_i$

After training, compute $H^l = P(h \mid v)$ using the final weights for layer l
End For

5 System architecture and integrated algorithmic workflow

The proposed hybrid architecture integrates rule-based learning, ensemble modelling, and deep representation learning within a unified framework. Traffic data from the original dataset are first processed by the Decision Tree module to extract dominant congestion rules, which are subsequently refined through Random Forest bagging to generate robust feature representations. These ensemble outputs form the visible layer of the Deep Belief Network, enabling hierarchical learning of nonlinear temporal and spatial traffic patterns across successive hidden layers as shown in Fig. 3.

A temporal and spatial dominance layer integrates real-time traffic deviations with learned historical patterns, allowing the framework to adapt to both recurrent and non-recurrent congestion events. The final output layer produces the predicted congestion state, supporting real-time traffic monitoring and decision-making as shown in Fig. 4.

This hierarchical design differentiates the proposed framework from existing hybrid models by explicitly structuring the information flow across decision, ensemble, and deep learning layers rather than combining them in parallel (as shown in Fig. 4).

While the preceding subsections describe individual components of the proposed framework, the complete end-to-end operational flow is formally summarized in Algorithm 4, which integrates the architectural design into a unified algorithmic workflow.

Algorithm 4 Integrated hybrid DT–RF–DBN traffic congestion prediction

1. Input:
   Original traffic dataset S,
   feature set F derived from S,
   temporal traffic observations X(t),
   Decision Tree depth d,
   number of trees T,
   learning rate η
2. Output:
   Predicted congestion state C(t)

1. Input Initialization
   Load the original traffic dataset S and extract the corresponding feature set F.
   Segment traffic observations into temporal windows X(t) as defined in Section 3.2.
2. Decision Tree Processing
   Apply Algorithm-1 (Tree_Creation) on dataset S using feature set F
   to obtain rule-based congestion decisions $B = \{B_1, B_2, \ldots, B_n\}$.
3. Random Forest Feature Aggregation
   Apply Algorithm-2 (Creation_RandomForest) with T trees on S.
   Aggregate ensemble outputs to generate intermediate representations
   $Z(x) = \{z1(x), z2(x), \ldots, zT(x)\}$, corresponding to the bagging layer.
4. DBN Visible Layer Construction
   Map the aggregated Random Forest outputs Z(x) to the visible layer (V_1–V_4)
   of the Deep Belief Network.
5. Hierarchical Representation Learning
   Train the Deep Belief Network using Algorithm-3 (DBN training)
   to learn multi-level hidden representations through successive hidden layers.
6. Temporal and Spatial Dominance Integration
   Integrate learned DBN representations with real-time deviation cues from X(t)
   in the temporal–spatial dominance layer to capture dynamic congestion behaviour.
7. Congestion State Prediction
   Propagate the integrated representation to the output layer
   to obtain the final congestion prediction C(t).
8. Return C(t)

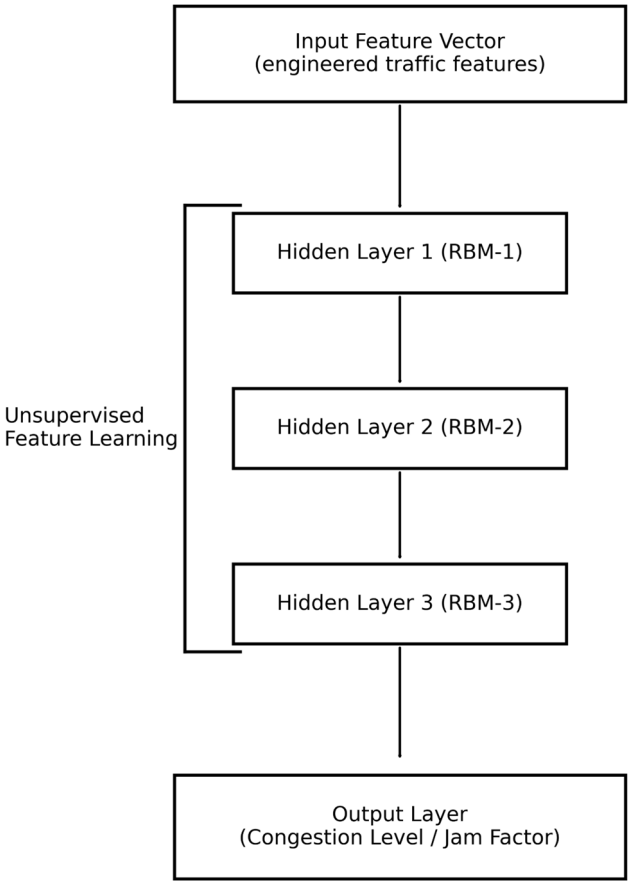


Fig. 3 Deep Belief Network architecture used for spatiotemporal traffic congestion prediction

The input to the proposed framework consists of temporally aggregated traffic feature vectors, including queuing-theoretic deviation indicators and engineered traffic attributes derived from traffic observations.

The output of the framework is the predicted traffic congestion state, represented as discrete congestion classes used for downstream traffic management and analysis.

5.1 Congestion state representation

Although the proposed framework processes continuous traffic variables such as average speed, vehicle density, queue length, and waiting time, the prediction task is formulated as a congestion state classification problem. Continuous traffic indicators are mapped to a finite set of congestion levels (e.g., low, medium, and high congestion) based on predefined threshold ranges derived from traffic flow characteristics. The model outputs the predicted congestion class corresponding to the observed traffic conditions, while numerical traffic values reported in the experimental analysis represent the underlying continuous indicators used for feature construction and evaluation.

5.2 Relationship between traffic indicators and congestion classification

In the proposed framework, traffic congestion classes are derived from a combination of continuous traffic indicators rather than from a single variable. Average speed, vehicle density, queue length, waiting time, and temporal context (e.g., time of day) jointly characterize the traffic state within each temporal window. Lower average speeds, higher vehicle

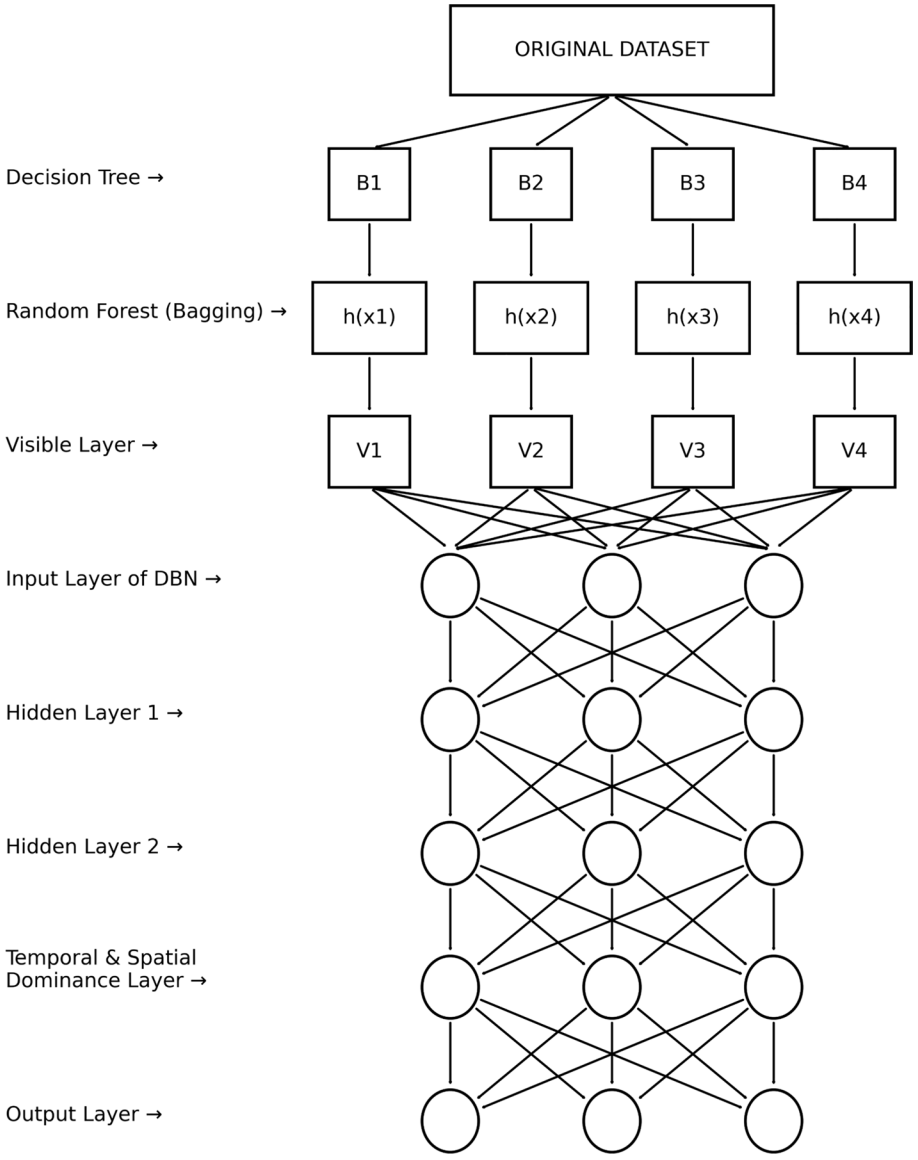


Fig. 4 Integrated framework combining decision tree, random forest, and deep belief network components for hierarchical traffic congestion. prediction

density, and increased queue-related measures indicate rising congestion severity, while temporal features help capture recurring peak and off-peak patterns. These indicators are evaluated together through rule-based decision logic and learned ensemble representations, and the resulting traffic state is mapped to discrete congestion levels (e.g., low, medium, high) using predefined threshold ranges. Thus, the classification output represents an interpretable abstraction of underlying numerical traffic conditions rather than a direct transformation of a single metric.

5.3 Decision logic for real-time. vs historical data dominance

The dominance between real-time and historical traffic information is determined using queuing-theoretic deviation measures computed over each temporal window. When the real-time deviation $HP(t) = h_p(t) - \widehat{h}_p(t)$ exceeds a predefined threshold θ, the framework prioritizes real-time features through the Decision Tree and Random Forest layers

for immediate congestion assessment. Conversely, when deviations remain within nominal bounds, historical traffic patterns aggregated over temporal windows are emphasized and processed through the Deep Belief Network to capture longer-term temporal dependencies. This threshold-based switching mechanism enables adaptive balancing between short-term responsiveness and long-range temporal modelling. The threshold θ is empirically selected using validation data to balance responsiveness and stability.

6 Experimental validation

6.1 Datasets

The proposed framework is validated on several real-world urban traffic datasets:

1. IBB Traffic Graph Dataset: Covers 2,451 sensors in Istanbul (urban and highway), with measurements at one-hour intervals over four years [5].
2. PeMS Dataset (California): Real-time and historical sensor data with over 39,000 detectors, including incident and weather information [6, 7].
3. HERE API (New Delhi): Combines traffic and weather data at five-minute intervals, including features such as jam factor, speed ratio, and humidity [18].

These datasets offer a diverse range of spatial and temporal resolutions, facilitating comprehensive evaluation.

6.2 Feature engineering and preprocessing

Feature engineering is performed to convert heterogeneous raw traffic data into a unified representation suitable for learning. Raw inputs such as traffic flow, average speed, vehicle density, temporal indicators (time of day, day of week), and queuing-theoretic deviation measures are aggregated over fixed temporal windows. These engineered features jointly capture spatial, temporal, and contextual traffic characteristics and are used as inputs to the decision tree and random forest modules. The aggregated feature representations are subsequently processed by the Deep Belief Network to learn higher-level temporal patterns.

Key preprocessing steps include:

1. Temporal Alignment: Synchronizing data from different sources (e.g., GPS, weather, sensor) into a unified time grid.
2. Spatial Mapping: Assigning GPS locations to specific road segments or intersections.
3. Feature Construction: Generating lagged variables, moving averages, and event indicators.
4. Missing Data Handling: Imputation using Random Forest or domain-specific heuristics.
5. Node Embedding: For graph-structured datasets, applying GLEE or Node2Vec to encode spatial relationships [5].

The feature set used in this study focuses on aggregate traffic variables that are consistently available across the evaluated datasets, including temporal indicators, vehicle density, speed, and queuing-theoretic measures. While lane-level infrastructure attributes can support more fine-grained congestion analysis, the objective of this work is network-level congestion prediction using commonly accessible aggregated traffic measurements. Incorporating lane-level traffic information is planned as a direction for future work when suitable datasets become available.

6.3 Validation pipeline implementation

The pipeline is implemented as follows:

1. Queuing Layer: Real-time computation of arrival, service, and queue metrics for each spatial node.
2. Decision Tree Layer: Rule-based initial classification of congestion likelihood based on engineered features.
3. Random Forest Layer: Ensemble learning for robust classification/regression and feature selection.
4. DBN Layer**:** Deep feature learning and temporal sequence modelling, supporting multi-step forecasts.

The model can operate in both centralized (batch learning) and decentralized (node-wise real-time prediction) modes, depending on the application context [6].

6.4 Comparative analysis

6.4.1 Baseline models

The proposed framework is benchmarked against several established models:

1. ARIMA and Statistical Models: For time-series prediction [8, 9].
2. Support Vector Regression (SVR): For regression-based congestion level estimation [6, 10].
3. CNN and RNN-based Deep Learning Models: For spatiotemporal sequence learning [7, 12, 13, 22].
4. Graph Neural Networks (GNN): For non-Euclidean spatial dependencies [5].

6.4.2 Baseline selection and comparison scope

Widely adopted statistical, machine learning, and deep learning approaches used in traffic congestion prediction are selected as the baseline models. Classical statistical models such as ARIMA are included to capture traditional time-series forecasting capabilities. Support Vector Regression represents regression-based learning approaches [39, 40].

Due to their interpretability, robustness, and frequent use in traffic analytics, Machine learning baselines such as Decision Tree and Random Forest are chosen Deep learning models. CNN- and RNN-based architectures, are considered to evaluate spatiotemporal representation learning performance [41].

Although recent graph neural network (GNN)–based models have demonstrated strong performance under dense sensor deployments, they typically require explicit graph construction, extensive spatial dependencies, and substantial computational resources. As the proposed framework emphasizes interpretability, modularity, and applicability under heterogeneous and data-sparse urban conditions, the comparative evaluation focuses on widely used and practically deployable baselines that align with these objectives.

6.4.3 Performance metrics

1. Precision and Recall: For congestion classification (normal vs. congested).
2. RMSE: For continuous congestion level prediction (e.g., jam factor).
3. F1-Score and AUROC: For overall classification quality.

4. Scalability and Real-Time Responsiveness: For decentralized vs. centralized architectures.

6.5 Model configuration and training protocol

All datasets were partitioned into training, validation, and test sets using a consistent split strategy across experiments to ensure fair comparison. Model hyperparameters and architectural depth were selected based on validation performance. The Deep Belief Network consists of multiple stacked RBM layers, with the number of layers chosen empirically to balance representation capacity and hyperparameters along with values are given in Table 4.

To ensure reproducibility, this section summarizes the configuration and training settings used for all evaluated models across datasets.

1. Decision Tree (DT): The Decision Tree model was implemented using the CART framework with Gini impurity as the split criterion. The maximum tree depth was constrained to prevent overfitting, and a minimum number of samples per leaf was enforced to improve generalization across datasets.
2. Random Forest (RF): The Random Forest model employed bootstrap aggregation with a fixed number of trees. Feature subsampling was applied at each split to enhance robustness, and the maximum tree depth and minimum samples per leaf were aligned with the Decision Tree configuration to ensure fair comparison.
3. Deep Belief Network (DBN): The DBN architecture consisted of multiple stacked Restricted Boltzmann Machine (RBM) layers followed by supervised fine-tuning. The number of hidden layers and hidden units per layer were selected empirically using validation data. A fixed learning rate was used during training, and unsupervised pre-training was performed prior to supervised optimization to stabilize convergence.
4. Data Partitioning Strategy: All datasets were partitioned using a temporal split to preserve chronological consistency and prevent information leakage. The earliest 70% of observations were used for training, the next 15% for validation, and the final 15% for testing. This strategy ensures that models are evaluated on future unseen data, reflecting realistic deployment conditions in traffic forecasting.

6.6 Experimental environment and reproducibility

To ensure full reproducibility of the experimental results, the computational environment, software versions, and configuration settings are explicitly documented.

All experiments were conducted in a Python-based environment using the following software stack:

- Python version: 3.12.7 | packaged by Anaconda, Inc. | (main, Oct 4 2024, 13:17:27) [MSC v.1929 64 bit (AMD64)]
- Scikit-learn version: 1.5.1
- NumPy version: 1.26.4

A fixed random seed was used across all experiments to guarantee deterministic behavior in model initialization and dataset partitioning.

The dataset was partitioned using a temporal split strategy to prevent information leakage. Specifically:

- 70% of the earliest observations were used for training
- 15% for validation
- 15% for testing

All model configurations, hyperparameter settings, and training scripts are provided in the Supplementary Materials to support independent replication of the experiments.

6.7 Fairness of model comparison

All comparative models were trained and evaluated using identical engineered input features, temporal observation windows, and a consistent temporal train–validation–test split to ensure fair benchmarking. Validation data were used under comparable search ranges for each model to train hyperparameters. This unified evaluation protocol isolates architectural differences as the primary source of performance variation and ensures that reported performance gains arise from model design rather than experimental inconsistencies.

It is important to examine their conceptual differences beyond raw predictive accuracy, in order to contextualize the role of the proposed framework relative to mainstream deep learning approaches. End-to-end data-driven learning of temporal or spatial dependencies are emphasized for modern traffic prediction models such as recurrent neural networks (LSTM/GRU) and graph neural networks (GNN) primarily emphasize. In contrast, the proposed framework adopts a hybrid hierarchical design. It integrates queueing-theoretic traffic indicators with interpretable ensemble learning and deep representation modeling. Table 5 summarizes the key architectural and methodological distinctions between these model families.

7 Results and discussion

7.1 Experimental objectives and hypotheses

The experimental study is designed to test three core hypotheses underlying the proposed hybrid framework. First, we hypothesize that integrating queuing-theoretic deviation modeling with hierarchical learning improves prediction accuracy compared with standalone statistical and machine learning models. Second, we hypothesize that

Table 4 Hyperparameter configuration of baseline models

Model	Hyperparameter	Value
Decision Tree	Criterion	Gini
	Max depth	Not constrained
	Min samples split	2
	Min samples leaf	1
	Random state	42
Random Forest	Number of trees	100
	Criterion	Gini
	Max depth	Not constrained
	Feature selection	sqrt
	Random state	42
Deep Belief Network	Hidden layers	[128, 64, 32]
	Activation	ReLU
	Learning rate	0.001
	Batch size	32
	Training epochs	50
	Dropout	0.2

the hierarchical architecture enhances robustness under heterogeneous traffic conditions and multimodal data inputs. Third, we hypothesize that consistent performance improvements across multiple real-world datasets indicate practical generalizability of the framework. The experiments in this section are structured to directly evaluate these hypotheses through cross-model comparison and multi-dataset validation.

7.2 Classification performance

The classification performance of the proposed framework was evaluated across multiple real-world datasets to assess predictive accuracy and cross-dataset consistency.

On the IBB Traffic Graph dataset, the proposed model (with feature engineering and node embedding) achieved:

1. Accuracy: Up to 95.3% (ExtraTrees with GLEE embedding)
2. Precision/Recall/F1: Above 90% across multiple classifiers (Random Forest, XGBoost, CatBoost)
3. AUROC: Consistently above 0.98, indicating excellent discrimination between normal and congested states [5].

These results indicate systematic performance improvements over standalone statistical and machine learning baselines. The gains can be attributed to the integration of queuing-theoretic deviation features, which improve responsiveness to short-term traffic fluctuations, and hierarchical learning, which captures nonlinear spatiotemporal dependencies. Ensemble stabilization through the Random Forest layer reduces prediction variance, while the DBN refines higher-level temporal representations.

On the New Delhi dataset, the framework achieved:

1. **RMSE**: An average value of 1.12 for one-week-ahead congestion prediction, outperforming baseline SVR and traditional time-series approaches.

The consistency of improvements across datasets with different sensing resolutions and traffic characteristics supports the robustness and generalizability of the proposed architecture. These findings empirically validate the hypothesis that combining interpretable traffic modeling with hierarchical learning yields more reliable congestion prediction than purely data-driven or purely analytical methods.

7.3 Comparative analysis

The proposed hybrid framework was compared with representative statistical, machine learning, and deep learning baselines to evaluate its performance under different modeling assumptions. The comparison examines predictive accuracy, robustness to heterogeneous data, and adaptability to dynamic traffic conditions.

7.3.1 The comparative results reveal three key observations

1. Improvement over classical statistical models

Traditional time-series approaches such as ARIMA have limited capacity to represent nonlinear and multimodal traffic behavior. By incorporating queuing-theoretic deviation features and hierarchical learning, the proposed framework responds more effectively to short-term congestion fluctuations. This leads to consistently lower prediction error and

higher classification accuracy than purely statistical baselines. This behavior is consistent with prior studies highlighting the limitations of purely statistical traffic prediction models in complex urban environments[25].

2. Advantage over standalone machine learning models

Standalone machine learning models such as Decision Trees and Random Forests provide interpretable predictions but struggle to capture higher-order spatiotemporal dependencies. The proposed framework integrates these models hierarchically with a DBN, transforming ensemble-stabilized features into richer temporal representations. This layered structure improves robustness to noise and missing data, resulting in more stable performance across datasets. Similar limitations of standalone machine learning approaches and the stabilizing benefits of ensemble strategies have been reported in earlier comparative traffic prediction studies [7].

3. Comparison with deep learning approaches

End-to-end deep learning models capture nonlinear traffic patterns but often require dense sensing infrastructure and large training datasets. The proposed framework achieves comparable predictive performance while preserving interpretability and deployment flexibility. By combining ensemble learning with hierarchical deep representation learning, the system improves accuracy without depending on fully end-to-end deep architectures. These findings align with existing work comparing hybrid and deep learning architectures for traffic prediction, which report similar trade-offs between accuracy and interpretability [38].

Taken together, the comparative results show that, the framework demonstrates consistent performance gains across all evaluated datasets rather than dataset-specific improvements. This cross-dataset consistency indicates that the architecture generalizes well to heterogeneous urban traffic environments.

From a practical traffic-management perspective, these gains reflect the complementary roles of the framework components. Queuing-theoretic modeling captures

Table 5 Conceptual comparison between mainstream deep learning models and the proposed hybrid framework

Aspect	LSTM / GRU models	GNN-based models	Proposed hybrid framework
Learning paradigm	End-to-end data-driven sequence modeling	End-to-end graph-structured deep learning	Hierarchical integration of theory-guided features + ensemble learning + deep representation learning
Interpretability	Low (black-box temporal representations)	Low (complex graph embeddings)	Moderate–High (queueing-theoretic indicators + interpretable tree ensembles)
Use of traffic domain knowledge	Implicit (learned from data)	Implicit via graph structure	Explicit integration of queueing-theoretic congestion indicators
Robustness to noisy / sparse data	Sensitive to missing or noisy inputs	Requires dense and well-structured sensor graphs	Improved robustness via ensemble decision layer and structured features
Computational complexity	Moderate–high (training recurrent networks)	High (graph construction + message passing)	Moderate (modular pipeline with separable components)
Deployment flexibility	Requires end-to-end retraining	Dependent on graph infrastructure	Modular and stage-wise deployable
Primary strength	Capturing temporal dependencies	Modeling spatial graph dependencies	Interpretable, theory-guided hierarchical learning with practical deployability

arrival–service dynamics at intersections and bottlenecks, while Decision Tree and Random Forest layers support interpretable real-time inference using traffic attributes such as vehicle density, speed, and incident conditions. The DBN extracts higher-level spatio-temporal representations from these stabilized features, enabling modeling of complex congestion evolution. This synergy explains the consistent empirical improvements and highlights reminder the framework's relevance for real-time traffic management.

7.4 Ablation studies

An ablation study was conducted to measure the effect of removing key components of the proposed framework.

1. Removal of feature engineering and node embedding

Without feature engineering and node embedding, classification accuracy dropped by up to 15% compared with the full model. This decrease indicates that these components contribute directly to prediction performance.

2. Replacement of the DBN

Replacing the DBN with shallow neural networks reduced multi-step prediction accuracy, particularly during irregular traffic events. The full model consistently produced lower prediction error than this simplified variant.

These results show that removing major components leads to measurable performance degradation, and the complete framework achieves the best overall accuracy.

7.5 Case studies

The case studies utilize the same external factors defined in Table 1, including weather conditions, temporal indicators, road quality, special events, and construction activities.

1. Incident-driven congestion

During a simulated road closure event in Istanbul, the model quickly detected abnormal queue formation via the queuing layer, triggered rule-based classification, and adapted DBN predictions based on real-time sensor inputs, outperforming static historical models in prediction accuracy.

2. Weather-influenced traffic surges

On the New Delhi dataset, the model successfully incorporated temperature, rainfall, and humidity features, yielding lower RMSE and higher classification accuracy than models that ignored weather inputs.

3. Decentralized real-time prediction

In the California PeMS dataset, decentralized node-wise predictors enabled each sensor to forecast its own congestion state using only local and neighbouring node data, demonstrating near real-time responsiveness and scalability.

7.6 Decision Tree performance and interpretability

The Decision Tree model achieved high accuracy in congestion classification with time-related features such as rush hours significantly influencing prediction performance. The

model demonstrated stable performance across both balanced and mildly imbalanced datasets.

A key advantage of the Decision Tree is interpretability, which allows direct examination of feature contributions and supports practical decision-making for traffic planning. However, the results also show sensitivity to tree depth, indicating that careful hyperparameter tuning is necessary to maintain consistent performance.

Figure 5 illustrates the feature importance distribution learned by the Decision Tree model. Average speed emerges as the dominant predictor, closely followed by vehicle density and queue-related indicators, while temporal and engineered features provide complementary contextual information. This distribution indicates that both real-time traffic dynamics and higher-level temporal patterns jointly influence congestion formation.

7.7 Comparative performance of prediction models

Figure 6 compares the performance of multiple models across accuracy, precision, recall, and F1-score. Traditional models show moderate performance, while machine-learning-based approaches achieve improved results. The proposed hybrid framework consistently outperforms all baselines, achieving scores in the range of 0.93–0.95 across all metrics. These results show consistent performance improvements across all evaluation metrics rather than improvement in a single criterion.

7.8 Discussion

7.8.1 Impact of queuing-theoretic features on congestion responsiveness

The experimental results demonstrate that integrating queuing-theoretic traffic state variables with hierarchical learning models improves responsiveness to non-recurrent congestion events such as accidents and weather-induced disruptions. By explicitly modelling traffic arrival and service dynamics, the framework adapts more effectively to sudden deviations from expected traffic patterns, supporting real-time traffic management and decision-making.

7.8.2 Role of hierarchical learning and interpretability

The inclusion of interpretable components, such as Decision Trees and Random Forests, enables transparent decision logic that allows traffic planners to understand how key features influence congestion prediction. At the same time, the Deep Belief Network captures nonlinear temporal dynamics, supporting short-term and multi-step congestion prediction without sacrificing interpretability at the system level.

7.8.3 Trade-off between interpretability and predictive performance

Compared to purely data-driven deep learning models, the proposed framework offers a balanced trade-off between interpretability, computational efficiency, and predictive performance. While graph-based and deep spatiotemporal models may achieve higher accuracy under dense sensor deployments, their practical adoption is often constrained by scalability requirements, limited interpretability, and deployment complexity [42].

By integrating queuing-theoretic deviation features with interpretable decision logic and hierarchical representation learning, the proposed approach maintains competitive prediction performance while improving transparency and adaptability. This balance

makes the framework particularly suitable for real-world traffic environments characterized by heterogeneous data availability and practical deployment constraints.

7.8.4 Practical implications, strengths, and limitations

From a practical perspective, the proposed hybrid framework is well suited for real-time urban traffic management scenarios where both predictive accuracy and interpretability are essential. The integration of queuing-theoretic traffic state deviations enables rapid adaptation to non-recurrent congestion events such as accidents, roadworks, and weather-related disruptions. Moreover, the use of interpretable models such as Decision Trees and Random Forests allows traffic authorities to understand and trust the decision logic, supporting practical deployment in intelligent transportation systems.

A major strength of the framework lies in its modular and hierarchical design. By decoupling theoretical traffic modelling from machine learning and deep representation learning, the proposed approach achieves a favourable balance between interpretability, scalability, and predictive performance. Unlike end-to-end deep learning models that require dense sensor networks and explicit graph construction, the framework operates effectively under heterogeneous data availability without relying on detailed road network graphs.

Despite these advantages, certain limitations remain. The Deep Belief Network component increases computational cost during training, which may restrict scalability in large-scale deployments. Additionally, queuing-theoretic approximations may not fully capture highly irregular traffic dynamics under extreme non-stationary conditions. Future work may address these limitations through adaptive learning strategies, model optimization, and edge or federated learning approaches.

7.8.5 Robustness to heterogeneous traffic conditions

The robustness of the proposed framework under heterogeneous traffic conditions is primarily achieved through its feature design and hierarchical learning structure. Variations in link lengths, roadway capacities, signal delays, and spillback effects are implicitly captured through queuing-theoretic traffic state variables such as arrival rate, service

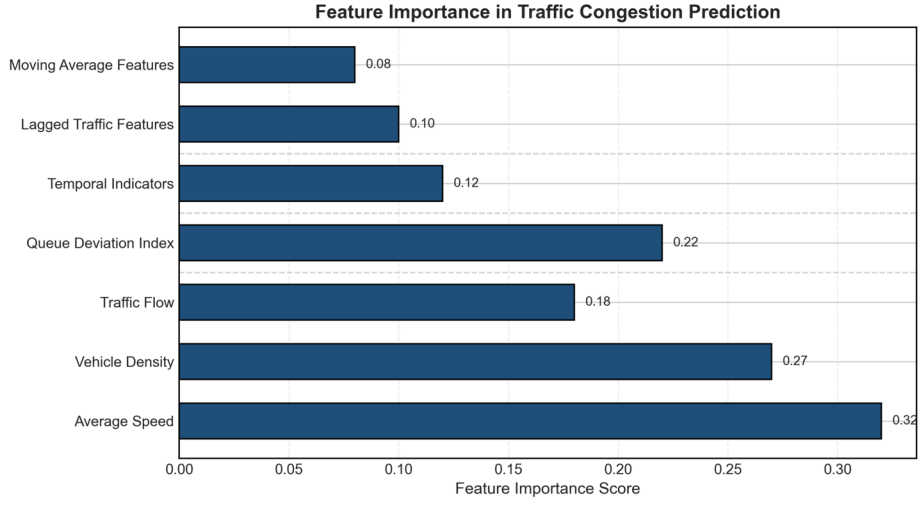


Fig. 5 Feature importance scores derived from the Decision Tree model for traffic congestion prediction. The grouped visualization highlights the relative contributions of engineered, temporal, queue-based, and traffic flow features to congestion classification

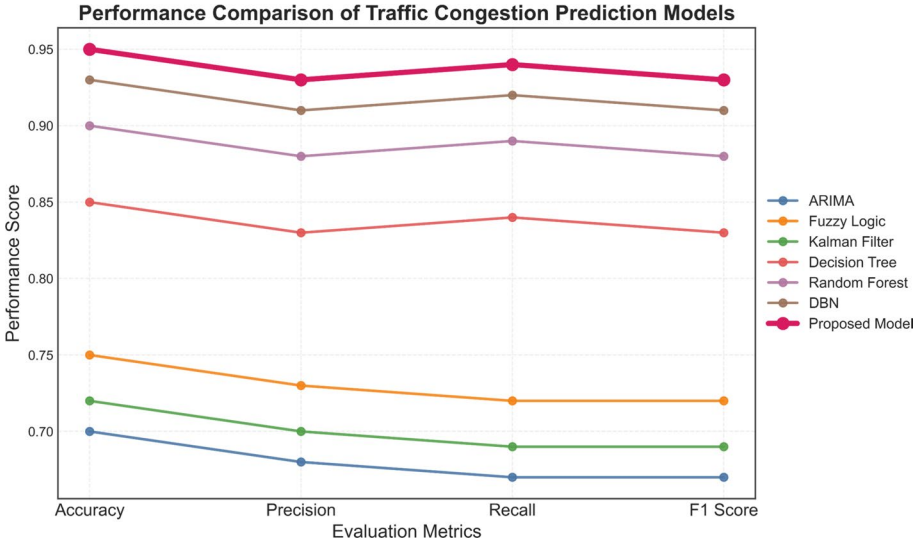


Fig. 6 Performance comparison of traffic congestion prediction models across accuracy, precision, recall, and F1-score

rate, queue length, and waiting time, which abstract local geometric and control differences into measurable congestion indicators. Since these variables reflect operational conditions rather than fixed network topology, the framework remains applicable across heterogeneous urban road segments.

Furthermore, the use of ensemble learning through Random Forests enhances robustness to variability and noise by aggregating decisions across multiple trees trained on different feature subsets and data partitions. Temporal segmentation and deviation-based features allow the model to adapt to signal-induced lost time and spillback interactions by capturing abrupt changes in service dynamics over successive time windows. While the framework does not explicitly model microscopic interactions at individual intersections, the experimental results indicate stable performance across datasets with diverse traffic characteristics, suggesting resilience to real-world heterogeneity commonly observed in urban networks.

7.8.6 Sensitivity to hyperparameters and data uncertainty

The sensitivity of the proposed framework to hyperparameter settings is mitigated by its modular and ensemble-based design. Decision Trees and Random Forests exhibit relatively stable behaviour across a range of depth, tree count, and sampling parameters, particularly due to ensemble averaging in Random Forests, which reduces variance and overfitting. For the Deep Belief Network, key hyperparameters such as the number of hidden layers, hidden units, and learning rate were selected using validation data, and empirical observations indicated stable performance within reasonable parameter ranges.

Regarding data uncertainty and incompleteness, the framework demonstrates resilience through both feature design and learning strategy. Queuing-theoretic traffic state variables abstract raw sensor noise into aggregated operational measures, reducing sensitivity to transient measurement errors. In addition, Random Forest–based learning provides robustness to missing or partially observed inputs, while temporal windowing enables the model to smooth short-term fluctuations caused by incomplete or delayed

data. Although extreme data sparsity or prolonged sensor outages may impact prediction quality, the results indicate that the framework maintains stable performance under realistic levels of uncertainty commonly observed in urban traffic sensing environments.

7.8.7 Handling non-stationary and sparse traffic data

Urban traffic data are inherently non-stationary due to temporal demand fluctuations, incidents, weather conditions, and evolving network dynamics. The proposed framework addresses non-stationarity through temporal segmentation and hierarchical learning. Traffic observations are processed within sliding temporal windows, enabling the model to adapt to short-term variations while preserving historical context. In addition, queuing-theoretic traffic state deviations explicitly capture departures from expected traffic behaviour, allowing rapid responsiveness to sudden regime shifts such as accidents or abnormal demand surges.

Sparse data scenarios are handled through feature abstraction and ensemble learning. Queuing-based metrics aggregate raw sensor measurements into higher-level traffic state representations, reducing reliance on dense sensor coverage. Random Forest–based learning further enhances robustness under sparse or partially missing inputs, while temporal aggregation mitigates the impact of intermittent observations. Although extreme sparsity may reduce predictive accuracy, the framework is designed to operate effectively under heterogeneous and uneven data availability commonly observed in real-world urban traffic systems.

8 Implementation challenges

Implementing a hybrid traffic prediction system involves several practical challenges that affect deployment and long-term operation.

One major issue is data collection and quality. Reliable prediction depends on accurate historical and real-time traffic data. In practice, traffic datasets often contain missing values, noise, and inconsistencies. Information is collected from multiple sources such as GPS sensors, cameras, and IoT devices, and these systems use different formats and resolutions. Combining such heterogeneous data into a unified framework requires careful preprocessing and synchronization. Another difficulty is the limited availability of labelled training data. Traffic conditions change frequently due to weather, road incidents, and human behaviour, which makes consistent data labelling difficult.

A second challenge relates to computational demands and real-time processing. Machine learning and deep learning models, especially architectures such as Deep Belief Networks (DBN), require substantial computing power during training. Processing large traffic datasets can be time-consuming and expensive. At the same time, real-time traffic prediction requires very low latency so that decisions can be made quickly. Cloud-based or high-performance computing platforms can support these requirements, but they introduce additional infrastructure costs and maintenance needs.

Another concern is model interpretability and reliability. Deep learning models often achieve strong predictive performance, yet their internal decision processes are not always transparent. Traffic authorities typically prefer systems that provide understandable explanations for their predictions. Hybrid frameworks that combine Decision Tree (DT), Random Forest (RF), and DBN components improve accuracy, but they also

increase system complexity. A practical implementation must balance predictive performance with interpretability so that users can trust the model's outputs.

Integration with existing traffic infrastructure presents further difficulties. Many urban traffic systems rely on legacy hardware and software that were not designed for AI-driven analytics. The hybrid prediction model must work alongside technologies such as smart traffic signals, navigation platforms, and roadside sensors. This requires standardized data formats and communication protocols. In addition, upgrading infrastructure to support advanced predictive systems can be costly and may require gradual implementation.

Security and privacy are equally important. Traffic datasets often include sensitive information about vehicle movement and travel behaviour. Protecting this data from unauthorized access is essential. Systems must follow privacy regulations and safeguard IoT devices against cyberattacks to prevent service disruption or data leaks [14, 16, 18].

From a deployment perspective, the computational load of the proposed framework is not evenly distributed. The Decision Tree and Random Forest layers have low inference latency and can operate efficiently at the edge, making them suitable for real-time applications. In contrast, the Deep Belief Network requires heavier computation during training but runs efficiently during inference. This component is therefore better suited for centralized or cloud environments. Such separation supports a flexible edge–cloud architecture, where lightweight modules handle immediate decisions while deeper learning tasks run centrally. A detailed evaluation of computational cost and latency across different hardware platforms remains an important area for future research.

9 Conclusion and future scope

9.1 Conclusion

Traffic congestion continues to be a critical challenge in modern and urban environment; thus, it is necessary to adapt frameworks which are intelligent and predictive. Our work proposes a hybrid traffic congestion prediction model integrating Decision Tree, Random Forest, and Deep Belief Network techniques which captures both interpretable traffic patterns as well as complex dynamics. After conducting various experimental evaluations on real world dataset, we can conclude that the proposed framework consistently outperforms traditional statistical approaches, supporting more accurate and reliable congestion prediction for traffic management applications.

Although the framework is effective, it faces challenges related to computational complexity and real-time scalability, mainly due to deep learning component. An important strategy for addressing these limitations is through model optimization and efficient deployment for further research. The primary contribution of this work lies in the structured integration of queuing-theoretic deviation modelling with hierarchical machine learning. It demonstrates that there is no need of sacrificing interpretability or scalability in order to achieve high predictive performance.

9.2 Future scope

Future research can focus on improving the scalability and real-time deployment of the proposed framework by incorporating edge computing and federated learning paradigms. The integration of IoT-enabled traffic infrastructure and connected vehicle data can further enhance the system's responsiveness to dynamic traffic conditions.

Additionally, advances in explainable artificial intelligence (XAI) can improve the transparency of deep learning components, increasing trust and usability for traffic authorities. Extending the framework to leverage multi-source traffic data and adaptive learning strategies will support the development of more efficient, sustainable, and intelligent urban traffic management systems.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s42452-026-08526-z.

Supplementary file1

Author contribution

A.Aditi Jha contributed with drafting of manuscript, integrating the ML models and analysing the implementation. B. Dr. R.S. Pandey has contributed with ideation and conceptualization C. Dr. Gyanendra Tiwary has worked for analysis and Literature Review D. Dr. Gaurav Londhe has worked forimplementation of the ML modeling.

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Data availability

The datasets used in this study are publicly accessible as follows: IBB Traffic Graph Dataset (Istanbul): https://github.com/erenolg/IBBTraffic•PeMS Dataset (California): https://zenodo.org/records/3939793•HERE Traffic API (Delhi): Accessible via the HERE Developer Platform (requires API key registration): https://developer.here.com/documentation/traffic

Declarations

Ethical approval

Not applicable.

Clinical trial number

Not applicable.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Competing Interests

The authors declare no competing interests.

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